

CS6101 PA2 Report

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October 23, 2025

1 Preprocessing steps

1.1 HotpotQA

- From the HotpotQA dataset, the question column contains the query, the context corresponding to the title in the supporting facts column refers to good documents, while the other contexts are the bad documents.
- All the contexts from HotpotQA were put in a Lucene index, which internally applies stemming on each word.
- The vocabulary was chosen by the following heuristics:-
 - The word is only alphabetical (only contains letter a-z and A-Z).
 - The corpus frequency is greater than 1.05 times the document frequency
 - The corpus frequency should be greater than 10.
 - The minimum word length should be 4.

On applying these conditions, our vocabulary size becomes **42654**.

1.2 WikiCLIR

- From the WikiCLIR dataset, the good documents are the ones for which are present in qrels table.
- Those queries are chosen which have atleast 10 and atmost 100 good documents.

2 Taskwise Metrics

The chosen hyperparameters for all tasks are:-

- $\lambda = 0.01$
- $margin = 0.2$
- $\tau = 0.05$
- $batch_size = 256$
- $lr = 10^{-4}$
- $epochs = 1$

2.1 Task 1

For BERT embeddings, average pooling was used.

Baseline Type	NDCG@10	MRR@10	Recall@100
BM25	0.2546	0.4982	0.4141
BERT	0.1233	0.2597	0.3526

Table 1: Results for Task 1 for different baseline types

2.2 Task 2

A is initialised as a tensor of shape (embed_dim, vocab_size) with values sampled from a Gaussian distribution $\mathcal{N}(0, 10^{-4})$. We used the Adam optimizer to obtain A.

For lower embed dims, the results are slightly better, more so in Triplet Loss.

All the results irrespective of the embed dim are much better than the BM25 and BERT baseline. We can see that with lower embed dim the results are slightly better and this improvement is more noticeable for the triplet loss.

Loss Function	Embed Dim	NDCG@10	MRR@10	Recall@100
Triplet Loss	100	0.2911	0.5814	0.4094
	300	0.2875	0.5748	0.4034
	700	0.2838	0.5668	0.3948
InfoNCE Loss	100	0.3150	0.6158	0.4416
	300	0.3146	0.6166	0.4407
	700	0.3143	0.6171	0.4390

Table 2: Results for Task 2 for different loss functions and embedding dims

2.3 Task 3

From WordNet, the synonym, hypernym and hyponym constraints were applied.

The results are much worse than the BM25 baseline but somewhat better than the BERT baseline except in Recall@100.

Loss Function	Embed Dim	NDCG@10	MRR@10	Recall@100
Triplet Loss	300	0.1386	0.3124	0.2639
InfoNCE Loss	300	0.1881	0.4108	0.3232

Table 3: Results for Task 3 for different loss functions

2.4 Task 4

For each token, the 10 most similar tokens were taken.

The results are worse than the BM25 baseline but somewhat better than the BERT baseline. The results are better than Task 3.

Loss Function	Embed Dim	NDCG@10	MRR@10	Recall@100
Triplet Loss	300	0.1897	0.3853	0.3628
InfoNCE Loss	300	0.2154	0.4367	0.3940

Table 4: Results for Task 4 for different loss functions